



Thinking CAPs

**Drive. Diagnose.
Verify. Contribute.**

Freestyle / HackFest | Eclipse SDV Hackathon 2026

October 6-8 2026



Existing needs. Reviewable contributions.

We advance project issues with working code, tests and reusable integration.

SUBMITTED

COM reliability

Repeated offer, stop, discovery and subscription integration tests.

[COM PR #1335](#)

SUBMITTED

S-CORE to SOVD

Expose diagnostic data resources through an OpenSOVD provider.

[Diagnostics PR #40](#)

LOCAL / VERIFIED

Diagnostic routing

Route UDS by logical address from DoIP to CAN targets.

[OpenBSW contribution](#)

Supporting work: lifecycle #704, SOME/IP #84 and native fault-storage persistence.

Measured quality, with explicit boundaries.

Native tests, source identities and review packets accompany the selected changes.

COMMUNICATION #1167

503 passed

6 skipped; inherited copyright findings retained.

LIFECYCLE #704

113 passed

Native cases for shared configuration generation.

OPENBSW ROUTER

46 unit tests

2 Bazel tests and 28 host integration tests passed.

Remaining gates: author/ECA where applicable, native CI, maintainer review and platform coverage.

[COM validation](#)

[Lifecycle #704](#)

[OpenBSW native gates](#)

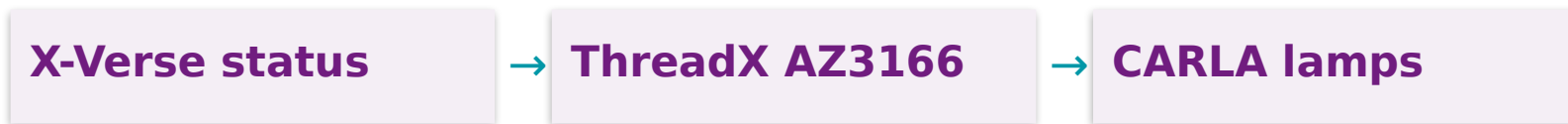
Connect projects through observable behavior.

Reusable integration paths link vehicle state, receiver diagnosis and embedded actuation.

RECEIVER / DIAGNOSIS



ZONAL LIGHTING



AutoSD: separately verified lighting deployment, native diagnosis and disturbance/recovery.

[Receiver campaign](#)

[Physical ThreadX](#)

[AutoSD guest evidence](#)

A contributor can pick up the next step.

Pinned inputs and portable records make the work understandable and repeatable.

01 / IDENTIFY

Pinned inputs

Source revisions,
dependencies,
configuration and
executable hashes.

02 / EXECUTE

Repeatable checks

Native regressions,
campaign verdicts and
owned teardown.

03 / CONTINUE

Explicit handover

Patches, PR links,
known gaps and
maintainer-facing next
steps.

Reference-host self-run demonstrated; independent operator and cold-machine reproduction pending.

[Reproduction guide](#)

[Contribution registry](#)

[AutoSD verification](#)

Useful to maintainers, integrators and testers.

Leave reviewable building blocks for the community to adopt and extend.

MAINTAINERS

Review and adopt

Bounded fixes and tests with recorded baselines and limitations.

INTEGRATORS

Reuse the pattern

Diagnostic adapters, vehicle contracts and deployment examples.

TEST ENGINEERS

Repeat the evidence

Fault campaigns, recovery checks and inspectable verdict artifacts.

Next: resolve native gates → obtain project review → repeat with another contributor.

[Maintainer PR](#)

[Integration handover](#)

[Router continuation](#)

Existing project work, moved forward.

Each selected contribution has an issue, an artifact and a concrete next action.

COM #1167

PR #1335

Integration coverage submitted. Next: CI and maintainer review.

DIAGNOSTICS #16

PR #40

Provider adapter submitted. Next: author/ECA and native validation.

LIFECYCLE #704

Verified patch

Configuration deduplication. Next: native upstream submission.

Prepared core: 4 Oct. Hardware and gateway evidence: 6–7 Oct. Attribution stays explicit.

[Existing COM issue](#)

[Existing diagnostics issue](#)

[Lifecycle #704](#)

Technology evidence, ready for inspection.

Three bonus technologies have meaningful runtime or testbench evidence.

openDuT / +0.10

Managed bench

Real receiver traffic,
communication
interruption and
recovery.

THREADX / +0.10

Actual RTOS

Linux simulation and
physical AZ3166
lighting-controller
evidence.

AUTOSD / +0.10

Real guest

Deployment, diagnosis,
managed disturbance
and reboot evidence.

Potential +0.30, subject to evaluation. Java/Jakarta is not claimed; final score is capped at 5.00.

[openDuT testbench](#)

[ThreadX hardware](#)

[AutoSD VM](#)



Built to be continued.

**Review the patches.
Run the evidence.**

github.com/Eclipse-SDV-Hackathon-Chapter-Four/Thinking_CAPs

October 6-8 2026



Evidence index for the technical interview.

Each link identifies a bounded contribution or a recorded integration result.

COM #1167 / native evidence

[Open source / evidence record](#)

OpenBSW router / native gates

[Open source / evidence record](#)

Dashboard / API and browser checks

[Open source / evidence record](#)

AutoSD / guest and source identities

[Open source / evidence record](#)

Lifecycle #704 / 113 cases

[Open source / evidence record](#)

Receiver / physical verdicts

[Open source / evidence record](#)

ThreadX / hardware and CARLA

[Open source / evidence record](#)

Fault storage / regression and patch

[Open source / evidence record](#)

Team, scope and template attribution.

Five contributors; clear ownership; preserved preparation and event-work boundaries.

Jefferson Nascimento — architecture, factory, ECU contributions

Bruno Campos — S-CORE application and X-Verse integration

Puru — OpenSOVD / CDA adapters and dashboard

Yasser — OpenSOVD, review and documentation

Siva — openDuT and integration-test evidence

Template: Eclipse Foundation AISBL and contributors / adapted under CC-BY-SA 4.0.

[Official slide template](#)

[Pitching-session guide](#)

[Freestyle scorecard](#)